Design Proposal: Mach-Zehnder Interferometer

FELIPE SOUZA SANCHES[1]

[1] *edx Phot1x - Username: FelipeSouzaSanches*
[*] *felipess37@gmail.com*

Abstract: This study investigates the performance of single-mode Silicon MZIs (500×220 nm) under fundamental TE polarization and TM. By implementing path-length imbalances (ΔL) of 40, 80, 160, and 250 μm, we demonstrate a systematic approach to extracting waveguide properties. The following sections present the MZI theory used in this study, the simulation parameters, and a detailed verification of the theory using experimental data extracted from fabricated devices.

1. Introduction

Modern optical communication is strongly linked with silicon photonics, being a field that is benefiting from the infrastructure of CMOS foundries that are becoming more and more mature and also capable of achieving a higher production volume [1].

It is seen that silicon photonics enables the integration of optical systems in single and also compact chips [2]. From the several existent photonics systems, the Mach-Zehnder Interferometer (MZI) is seen as an indispensable component, mainly due to it versatility, to be used from signal modulation to wavelength filtering [3].

The current work details the design of several MZI configurations, targetting to measure waveguide group indices using an experimental setup. These designs are optimized for E-beam lithography to guarantee reliable fabrication. In this work, it is provided an overview of the MZI theory, component-level modeling, and the design variations.

2. Theory

In silicon photonics, it is possible to find several types of fundamental blocks. In this study, it will be introduced a Mach-Zehnder interferometer with 2 Y-branches, as shown in Fig. 1.

The MZI block operates by splitting the light in 3-dB, using a Y-branch, into two waveguides and then bringing them together into a single output, by using again another Y-branch. By making the lengths of the waveguides (being the branches of the interferometer) having different values, such as L_1 and L_2, we make this system unbalanced, triggering the interference of the field distribution, in the system's waveguide.

The expected output, observed in MZI systems are sinusoidal variations caused by the length imbalance, defined as free spectral range. The mathematical deduction will be done in the following sessions [3].

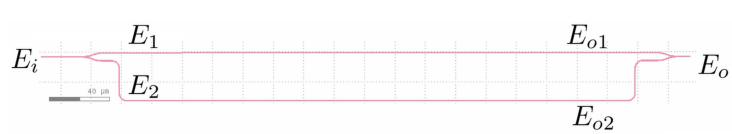


Fig. 1. Conceptual setup of an unbalanced Mach-Zehnder interferometer.

2.1. Theoretical Framework

This section introduces the basic equations to model a simple MZI. Equation 1 shows the propagation of light inside the waveguide.

$$\beta_{1,2} = \frac{2\pi n_{1,2}}{\lambda} \tag{1}$$

As observed in 1, at the end of each branches, the fields are described by equation 2 and 3. In these expressions, α_1 and α_2 represent the power attenuation coefficients.

$$E_{o1} = \frac{E_i}{\sqrt{2}} \exp\left(-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1\right) \tag{2}$$

$$E_{o2} = \frac{E_i}{\sqrt{2}} \exp\left(-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2\right) \tag{3}$$

At the second Y-junction, on its output, the total output field E_o can be described as:

$$E_o = \frac{1}{\sqrt{2}}(E_{o1} + E_{o2}) = \frac{E_i}{2}\left[\exp\left(-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1\right) + \exp\left(-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2\right)\right] \tag{4}$$

Since the optical intensity I is proportional to the square of the electric field magnitude ($I \propto |E|^2$), the output intensity I_o is given by:

$$I_o = \frac{I_i}{4}\left|\exp\left(-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1\right) + \exp\left(-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2\right)\right|^2 \tag{5}$$

It is possible to simplify this model by assuming nearly lossless waveguides ($\alpha_1 = \alpha_2 \approx 0$), and given that the waveguides in both branches are structurally identical, it results in the same effective index ($n_1 = n_2 = n_{\text{eff}}$) and propagation constant ($\beta_1 = \beta_2 = \beta$).

As a result, the interference found on this model, it is just linked to the physical path length difference, $\Delta L = L_2 - L_1$, as despicted in equation 6.

$$I_o = \frac{I_i}{2}\left[1 + \cos(\beta \Delta L)\right] \tag{6}$$

Another way to understand equation 6 is through the normalized transfer function:

$$\frac{I_{\text{out}}}{I_{\text{in}}} = \cos^2\left(\frac{\Delta\phi}{2}\right) \tag{7}$$

where the total phase difference $\Delta\phi$ is defined as:

$$\Delta\phi = \beta \Delta L = \frac{2\pi n_{\text{eff}}}{\lambda} \Delta L \tag{8}$$

The periodic nature of Eq. 7 results in a transmission spectrum characterized by distinct interference fringes. The waveguide's group can be extracted by using thoses spectral peaks and nulls [3].

2.2. Free Spectral Range (FSR) and Group Index Extraction

As described at the beginning of this session, from the sinosoidal functions, observed in the output of the MZI system, it is possible to extract the FSR (Free Spectral Range), a key metric used for characterizing the device's spectral response.

For an imbalanced MZI, the FSR is a function of both the physical path imbalance (ΔL) and the waveguide group index (n_g) [3], as seen in equation 9.

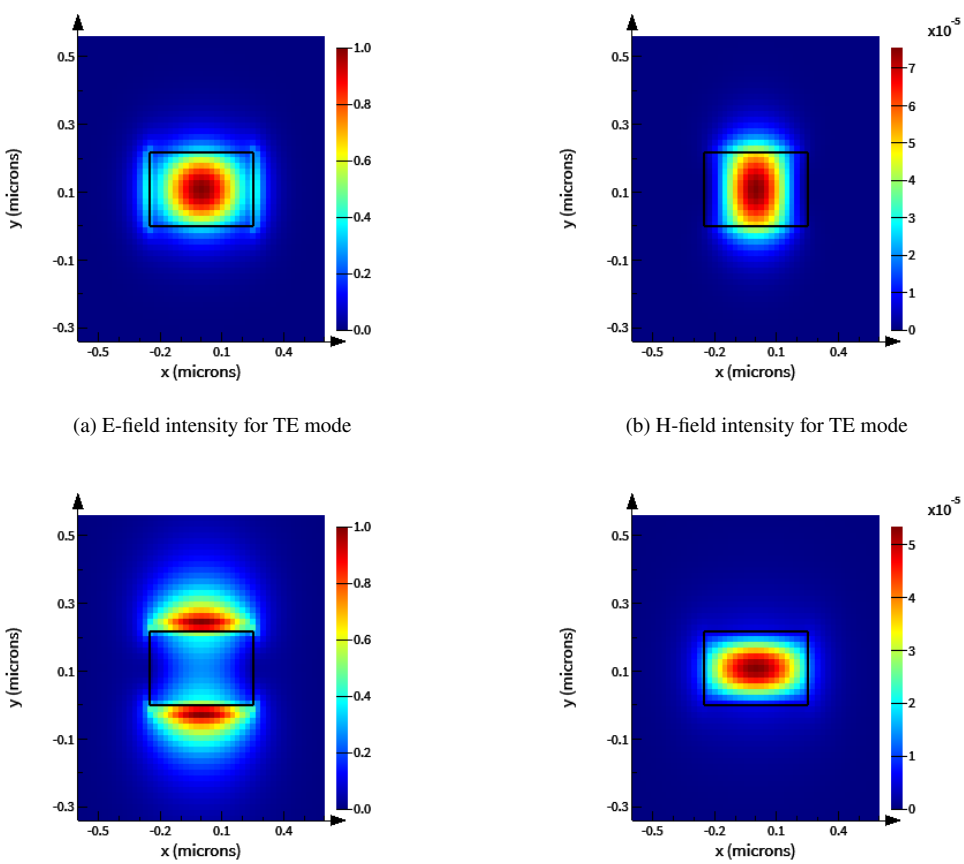
$$FSR = \frac{\lambda^2}{n_g \Delta L} \tag{9}$$

It is important to highlight that n_g is more important than the standard refractive index (n_{eff}) because the FSR depends on how the entire pulse is delayed between the two branches, in our MZI. If instead n_{eff} is used to calculate the Free Spectral Range, at the end the results would be incorrect. Due to the dispersive behavior in silicon, the phase velocity suffer fast variations that it "pulls" or "pushes" the group envelope, resulting in changes in the effective spaces between the observed interference peaks. [3].

3. Modelling and Simulation

3.1. Waveguide Design and Modal Analysis

The devices use a standard **strip waveguide** architecture. It is used a fixed **220 nm** silicon thickness and a waveguide width of **500 nm**. This geometry is a standard industry benchmark for the C-band, as it has high modal confinement and also supports single-mode operation for the fundamental **quasi-TE polarization** and **quasi-TM polarization**, shown in Fig. 2.



(a) E-field intensity for TE mode

(b) H-field intensity for TE mode

(c) E-field intensity for TM mode

(d) H-field intensity for TM mode

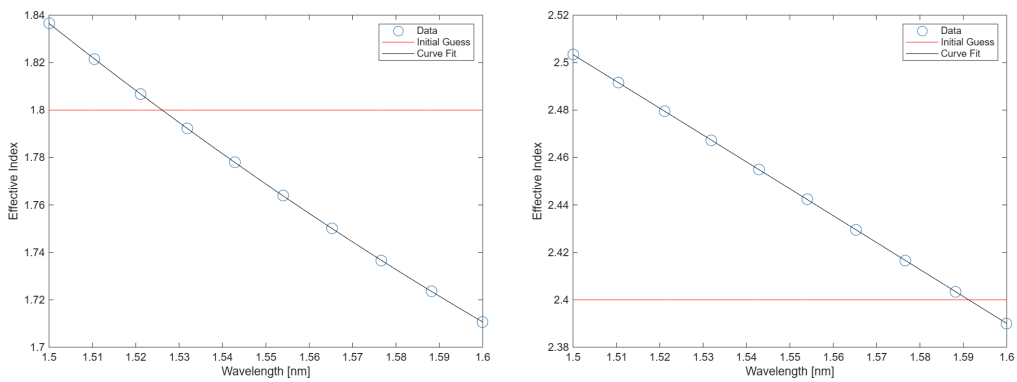
Fig. 2. Mode profiles obtained by using eigensolver analysis [4], for a silicon strip waveguide of 500×220 nm.

The simulation also shows the optical confinement, achieved by the silicon core, for both TE and TM modes.The quasi-TE mode shows higher confinement inside the silicon core, however, the quasi-TM mode has a larger field dispersion extending into the cladding, as presented in Fig. 2a and Fig. 2c, respectively.

To accurately characterize the modal properties of the given geometry, numerical simulations are performed using an eigenmode solver [4], for the given waveguide cross-section. From that, it is possible to extract of the effective index and dispersion characteristics, with higher accuracy, for the MZI designs. For TE mode the effective index obtained is $n_{eff} = 2.447$ and for TM mode, the effective index obtained is $n_{eff} = 1.769$.

Based on the second-order Taylor expansion, the effective index can be represented by Eq. 10.

$$n_{\text{eff}}(\lambda) = n_{\text{eff1}} + n_{\text{eff2}}(\lambda - 1.55) + n_{\text{eff3}}(\lambda - 1.55)^2 \tag{10}$$



(a) Effective index for TE mode is given by $n_{\text{eff}}(\lambda) = 2.447 + -1.133(\lambda - 1.55) + -0.044(\lambda - 1.55)^2$

(b) Effective index for TM mode is given by $n_{\text{eff}}(\lambda) = 1.769 + -1.259(\lambda - 1.55) + 1.913(\lambda - 1.55)^2$

Fig. 3. Curve fitting for effective index in function of wavelength.

Table 1 shows all coefficients obtained after running the numerical simulation, for both TE and TM. Fig. 5 display the curve fitting using the coefficients.

Table 1. Second-order Taylor coefficients for the effective index (n_{eff}) of TE and TM modes at $\lambda_0 = 1.55$ µm with 6-decimal precision.

Mode	n_{eff1}	n_{eff2} (µm^{-1})	n_{eff3} (µm^{-2})
Quasi-TE	2.446821	-1.133395	-0.043934
Quasi-TM	1.768821	-1.258670	1.913483

3.2. Circuit Simulation Setup

The circuit was modeled using the Lumerical INTERCONNECT circuit simulator, with the configuration depicted in Fig. 4. This system-level simulation uses the dispersive waveguide properties calculated before, and also S-parameter models for the grating couplers and Y-branch elements, also included in the simulation. This ensure high accuracy to predict the spectral behavior of the device that will be fabricated.

The spectral response of the input and output grating couplers is shown in Fig. 5a. The data show a Gaussian-like profile, and its peak transmission is centered around 1550 nm, being the target C-band wavelength range for this MZI project.

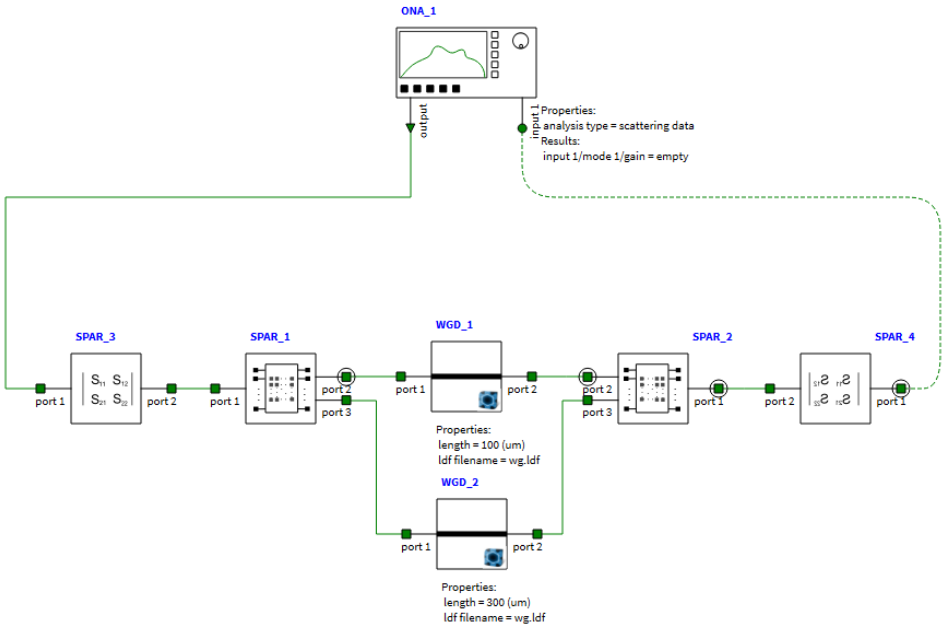
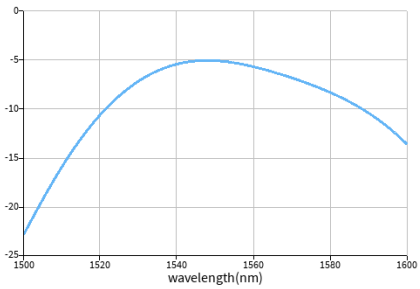
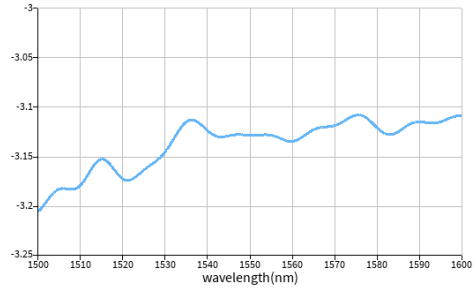


Fig. 4. System-level circuit created on Lumerical INTERCONNECT environment. The schematic illustrates the integrated simulation, in which waveguide segments are connected with S-parameters of the Y-junctions (named as SPAR_1 and SPAR_2) and grating couplers (named as SPAR_3 and SPAR_4), to obtain the full spectral response of the MZI.



(a) Simulated transmission characteristics for a back-to-back grating coupler.



(b) Simulated transmission spectrum of the Y-branch splitter, demonstrating broadband 3-dB power splitting.

Fig. 5. Spectral response from 1500 to 1600 nm.

The spectral response for the Y-branch splitter is presented in Fig. 5b. It is possible to see a stable loss of approximately 3 dB across the C-band, showing that the input power is equally divided betzeen the 2 outputs, with minimal parasitic loss.

3.3. MZI Design Variations

The designs for the MZIs was executed on KLayout software [5]. Some classical MZIs with different length imbalances were implemented. Besides that, a constant waveguide cross-section and identical Y-junctions were used in the designs, and TE and TM polarizations were also

explored. Fig. 6 shows the first layout draft with 3 different MZIs.

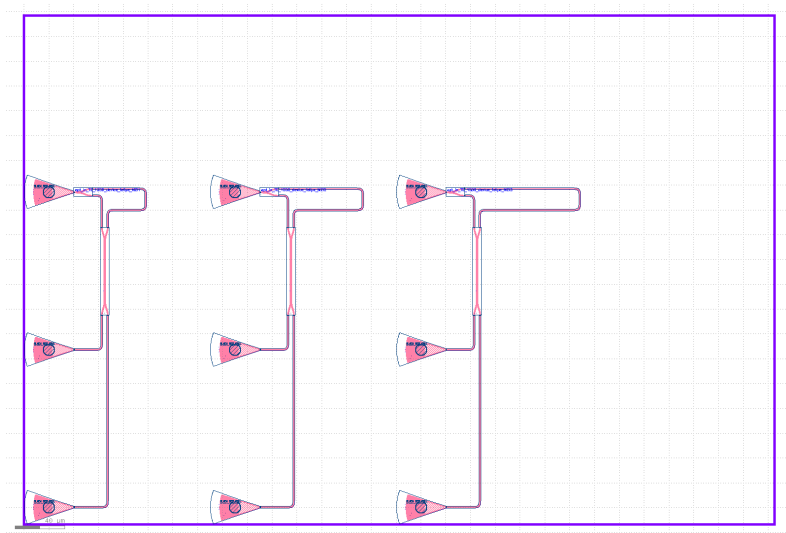


Fig. 6. First draft setup, using KLayout mask design for the three MZI variants.

The final layout is shown in Fig. 7. Table 2 summarizes the geometric parameters for each design, with its respective predicted FSR, calculated at a center wavelength of 1550 nm. For designs using TE mode, the simulated group index is $n_g \approx 4.197$, whereas for TM mode, the group index is $n_g \approx 3.75$.

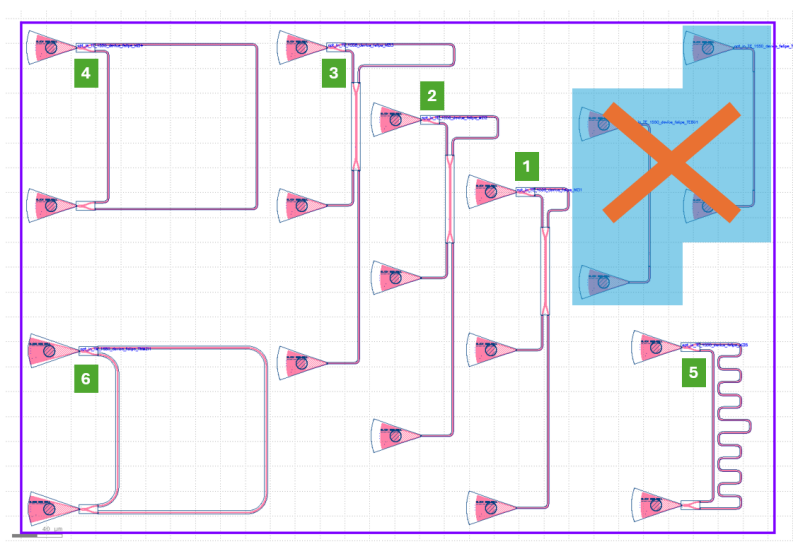


Fig. 7. Final KLayout mask designs for 6 MZI variants. Each design used in this study is numbered, and the blue region shows layouts that will not be taken into consideration, as they are not an MZI design.

The designs 4, 5, and 6 have the same ΔL. The variation to be studied are the waveguide serpentines influence, by making comparisons between designs 4 and 5, and a comparison between TE and TM modes, by analyzing the result of designs 4 and 6.

Table 2. MZI Design Parameters and Predicted Free Spectral Range

Design #	Width	Mode	L_1 (µm)	L_2 (µm)	ΔL (µm)	Expected FSR (nm)
1	500 nm	TE	30	70	40	14.30
2	500 nm	TE	30	110	80	7.15
3	500 nm	TE	30	190	160	3.58
4	500 nm	TE	140	390	250	2.29
5	500 nm	TE	140	390 (waveguide serpentines)	250	2.29
6	500 nm	TM	140	390	250	2.56

4. Fabrication

The photonic integrated circuits (PICs) were fabricated using the NanoSOI MPW process, done by Applied Nanotools Inc. [6] The technical specifications and process used in this project are summarized below:

Substrate and Material: The devices were processed on 200 mm SOI wafers. Besides that, the silicon layer height is 220 nm, and it is buried in a oxide (BOX) layer of 2 µm. Later, the wafer is diced into 25 × 25 mm substrates.

Electron Beam Lithography (EBL): The patterns were created by a JEOL JBX-8100FS 100 keV EBL system.

Etching and Development: Once the exposure is complete, the patterns are developed using a solution called tetramethylammonium sulfate (or TMAH). The different shapes of the circuit are then etched into the silicon by an isotropic ICP-RIE (Inductively Coupled Plasma Reactive Ion Etching).

Cladding and Metrology: At the end, a 2.2 µm thick SiO_2 upper cladding is deposited using plasma-enhanced chemical vapor deposition (or PECVD) at 300°C. After, it is verified the layer thicknesses and etch quality by using mainly the Scanning Electron Microscopy (SEM) method.

However, the fabrication process can cause a mismatch among devices, generating variability in both the width and height of the waveguide. This work uses, as reference, the variation provided by Soitec, France.

- Wafer diameter: 6";
- Si height: average size of 219.2nm, with a σ deviation of ± 3.9nm;
- Si width: 500 +10/-30nm;

Those information are important to execute the corner analysis of the optical device. Besides the nominal dimension, it is also simulated 4 different corners, as shown in Fig. 8, using the deviations for the waveguide.

Based on the corners diamensions for the device's waveguide, it was possible to obtain the effective index (n_{eff}), and the group index (n_g) at λ_0 = 1550 nm, as seen in Tables 3 and 4.

For each MZI layout, it was also calculated the predicted FSR, also considering the corner variations. Table 5 display the maximum and minimum values expected to be found in the measurements, obtained from the fabricated devices.

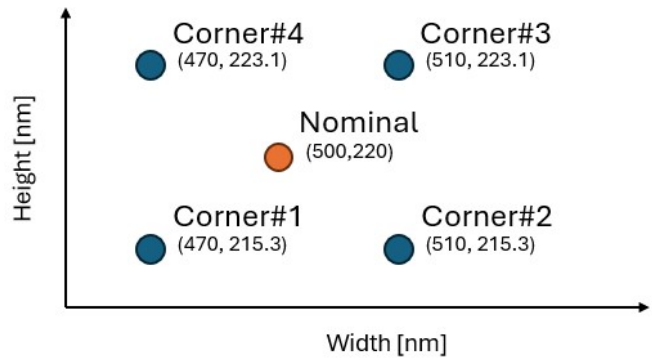


Fig. 8. Corner analysis representation, displaying variation in both width and height, from the waveguide.

Table 3. Corner Analysis for **TE Mode**: Effective Index (n_{eff}), Polynomial Models, and Group Index (n_g) at λ_0 = 1550 nm.

Corner	n_{eff} **at** 1.55 **µm**	**Taylor Expansion Polynomial** ($n_{\text{eff}}(\lambda)$)	n_g
Nominal	2.4468	$2.4468 - 1.1334(\lambda - 1.55) - 0.0439(\lambda - 1.55)^2$	4.2036
Corner 1	2.3725	$2.3725 - 1.2132(\lambda - 1.55) - 0.0341(\lambda - 1.55)^2$	4.2529
Corner 2	2.4416	$2.4416 - 1.1213(\lambda - 1.55) - 0.0333(\lambda - 1.55)^2$	4.1796
Corner 3	2.4722	$2.4722 - 1.1068(\lambda - 1.55) - 0.0465(\lambda - 1.55)^2$	4.1878
Corner 4	2.4039	$2.4039 - 1.1989(\lambda - 1.55) - 0.0528(\lambda - 1.55)^2$	4.2621

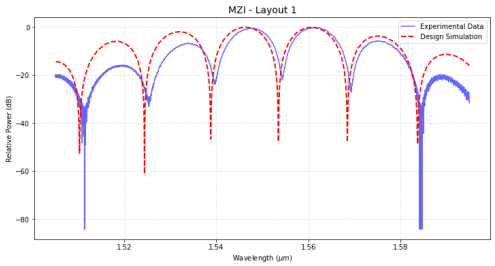
Table 4. Corner Analysis for **TM Mode**: Effective Index (n_{eff}), Polynomial Models, and Group Index (n_g) at λ_0 = 1550 nm.

Corner	n_{eff} **at** 1.55 **µm**	**Taylor Expansion Polynomial** ($n_{\text{eff}}(\lambda)$)	n_g
Nominal	1.7688	$1.7688 - 1.2587(\lambda - 1.55) + 1.9135(\lambda - 1.55)^2$	3.7198
Corner 1	1.7167	$1.7167 - 1.1390(\lambda - 1.55) + 2.0469(\lambda - 1.55)^2$	3.4821
Corner 2	1.7443	$1.7443 - 1.1889(\lambda - 1.55) + 1.9518(\lambda - 1.55)^2$	3.5870
Corner 3	1.7963	$1.7963 - 1.3151(\lambda - 1.55) + 1.8184(\lambda - 1.55)^2$	3.8346
Corner 4	1.7664	$1.7664 - 1.2747(\lambda - 1.55) + 1.9529(\lambda - 1.55)^2$	3.7422

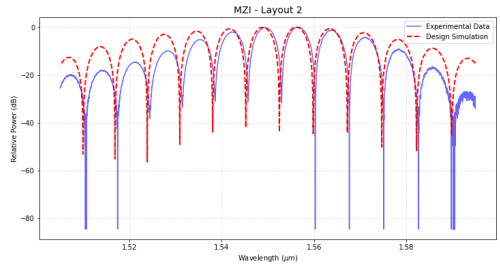
Table 5. Predicted FSR (nm) across Manufacturing Corners for MZI Designs 1–6 (Min: Corner 2, Max: Corner 4).

Design #	ΔL (µm)	Mode	Nominal FSR	Min FSR (C2)	Max FSR (C4)	Variation (ΔFSR)
1	40	TE	14.289	14.092	14.370	0.278
2	80	TE	7.144	7.046	7.185	0.139
3	160	TE	3.572	3.523	3.593	0.070
4	250	TE	2.286	2.255	2.299	0.044
5	250	TE	2.286	2.255	2.299	0.044
6	250	TM	2.583	2.506	2.760	0.254

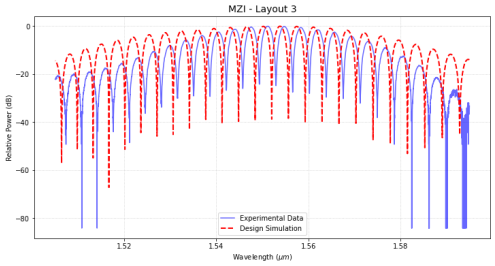
5. Experiment data



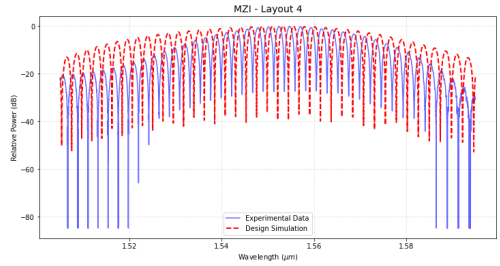
(a) Design 1, MZI with TE mode with ΔL = 40 µm.



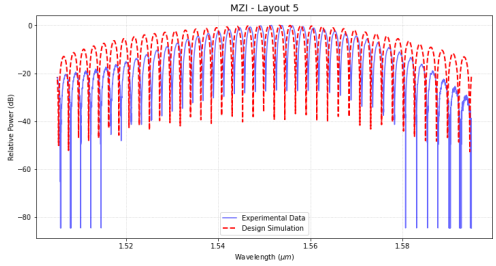
(b) Design 2, MZI with TE mode with ΔL = 80 µm.



(c) Design 3, MZI with TE mode with ΔL = 160 µm.



(d) Design 4, MZI with TE mode with ΔL = 250 µm.



(e) Design 5 (waveguide serpentines), MZI with TE mode with ΔL = 250 µm.

Fig. 9. Overlay of experimental data and simulated transfer functions for all MZIs designed, from 1 to 5.

The fabricated devices were characterized using an automated test station, described in [7, 8], and the measurement process was controlled by a software created in Python [9].

The setup is composed by an **Agilent 81600B** tunable laser, being the optical input source, and also an **Agilent 81635A** optical power sensor, as output detectors. Besides that, it is used a laser that sweeps its wavelength from 1500 nm to 1600 nm, in a resolution of 10 pm. The polarization control is done by a **Polarization Maintaining (PM) fiber array** [10].

For the TE mode, light was coupled directly into the grating couplers, whereas for TM mode characterization, a 90° rotation was applied to the input fiber to align the polarization with the TM-specific grating couplers, as seen in [11].

Thus, for this study, it was executed the characterization of six Mach-Zehnder Interferometer (MZI) layouts, using the setup described before.

Optical transmission spectra were collected for five TE mode designs (Layouts 1–5), as shown in Fig. 9, which presents different lengths (ΔL) ranging from 40 µm to 250 µm.

The raw test data for the TE designs showed very clear interference patterns, although the noise level around the values of 1500 nm and 1600 nm was evidently higher, the curve fitting was still possible to be performed accurately. Besides that, the periodicity of both simulated and experimental waveforms were similar.

Layout 6 was designed for TM mode operation. However, due to a fabrication error it was not possible to perform any measurement, and consequently, the interference patterns for Layout 6 could not be analyzed. Thus, all experimental data provided in this study are only for TE-polarized designs.

6. Analysis

The analysis done in this study is focused on the comparison between extracted experimental parameters and the nominal and corner simulations (C1, C2, C3, and C4), to finally determine the MZIs fabrication quality and model accuracy.

The extracted effective index (n_{eff}) and group index (n_g) were plotted for each design separately, as seen in Fig. 10-14, (a) and (b). Fig. 15 and Fig. 16 summarize all design results, and also show, respectively, experimental n_{eff} and n_g delimited by simulation corners, and display nominal values from simulation as reference.

The experimental values for n_{eff} at 1.55 µm remained consistently, from 2.42 to 2.44. This is inside the limits obtained by the corners C1 (lower limit) and C3 (upper limit). Thus, it is possible to validate the phase-velocity model of the waveguide used in this study.

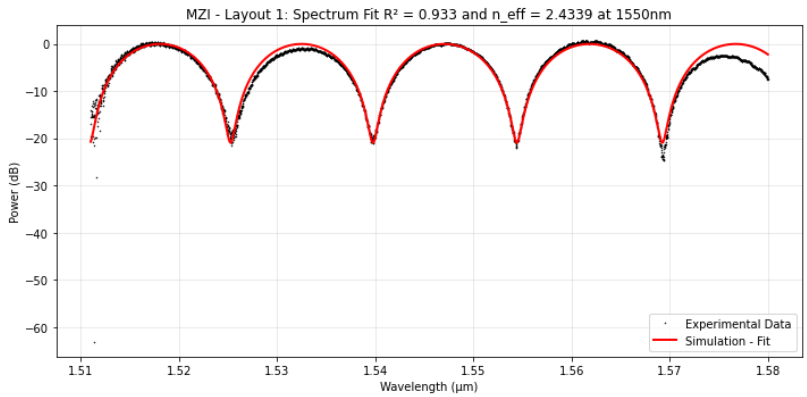
The group index values ended up being a bit lower than the nominal simulation value of 4.20. Only MZI designs 4 and 5 were inside the proposed limits. Since n_g is highly sensitive to the waveguide cross-section, this shows a small deviation in the group velocity. The results and the Taylor expansion are displayed separately for each design variation in Fig. 10-14.

The FSR is also a very important metric for verifying the fabricated MZI, and its physical integrity and quality. Fig. 10-14(c) show the to autocorrelation lots to determine the FSR, and Table 6 displays all FSR for each design variation, with their corner variation values defining the upper and lower limits.

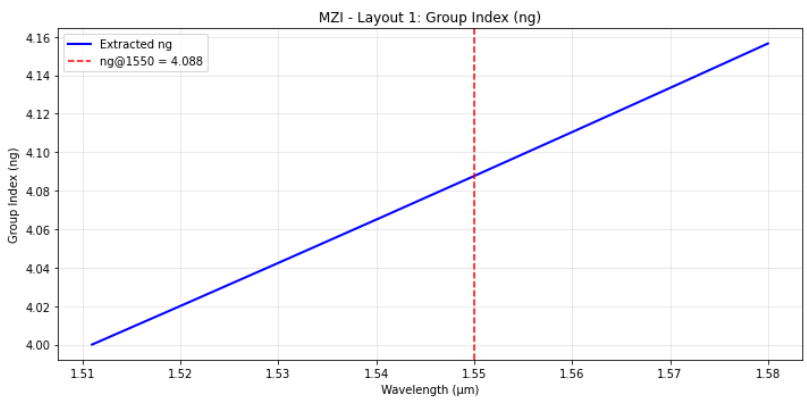
Besides that, the comparison between the results obtained from Layout 4 (straight) and Layout 5 (serpentine) for a fixed $\Delta L = 250$ µm does not present any relevant variation. The Experimental FSR error values are separated by only 0.3% from each other, confirming that the serpentine folding does not add any dispersion or deviations in the group index.

7. Conclusion

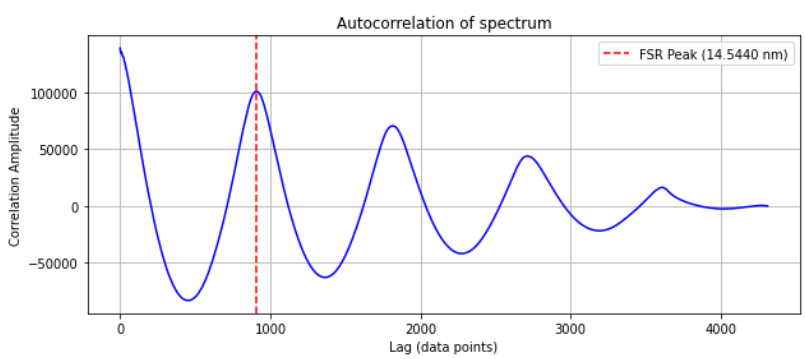
During this study, it was possible to validate the theoretical design and simulation models for TE-polarized, and also, to characterize the Mach-Zehnder Interferometer (MZI) layouts 1 to 5 successfully.



(a) Experimental data vs. fitted MZI transfer function for Layout 1.

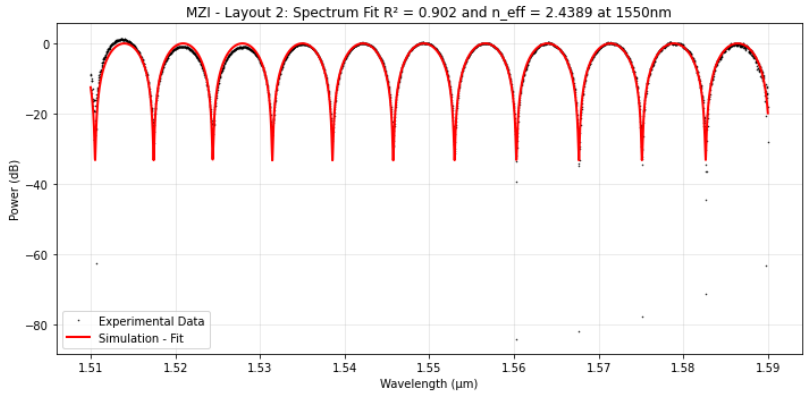


(b) Extracted group index n_g for Layout 1.

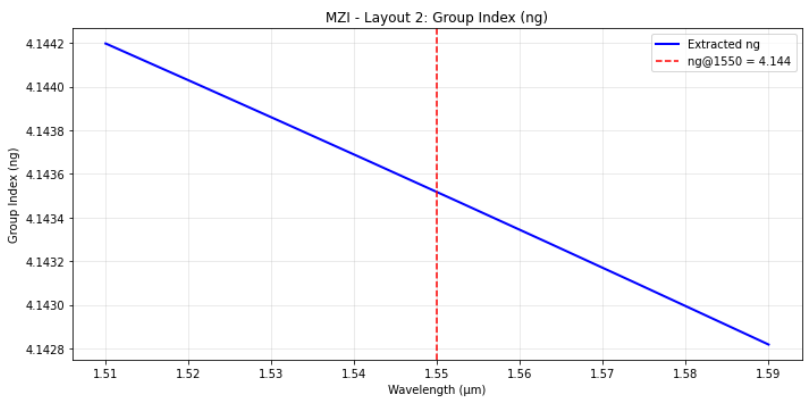


(c) Autocorrelation for Layout 1 used to determine initial FSR.

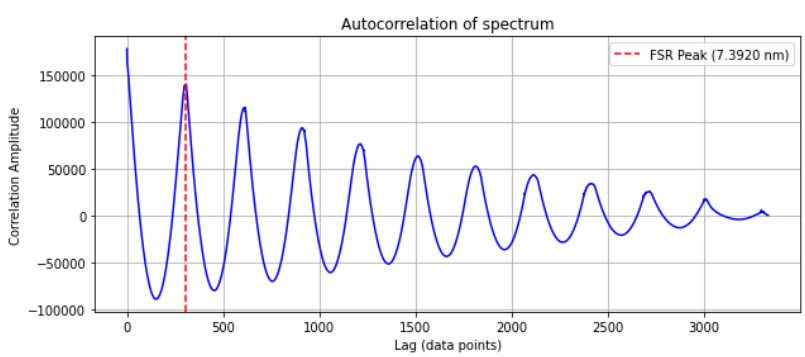
Fig. 10. Full analysis of MZI Layout 1. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4339$, $n_2 = -1.0669\,\mu m^{-1}$, and $n_3 = -0.7337\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.



(a) Experimental data vs. fitted MZI transfer function for Layout 2.

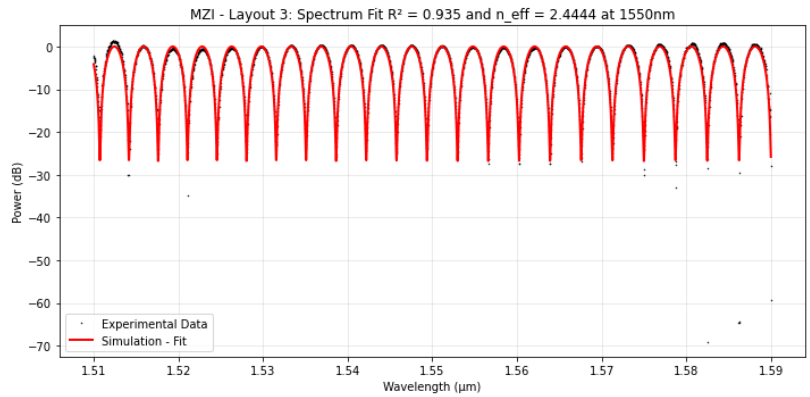


(b) Extracted group index n_g for Layout 2.

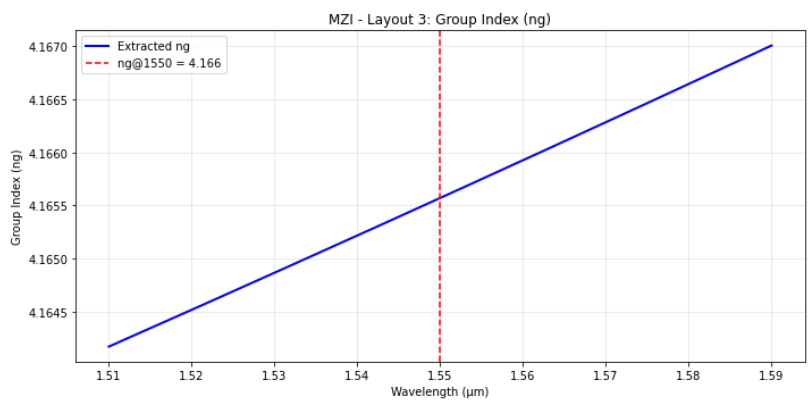


(c) Autocorrelation for Layout 2 used to determine initial FSR.

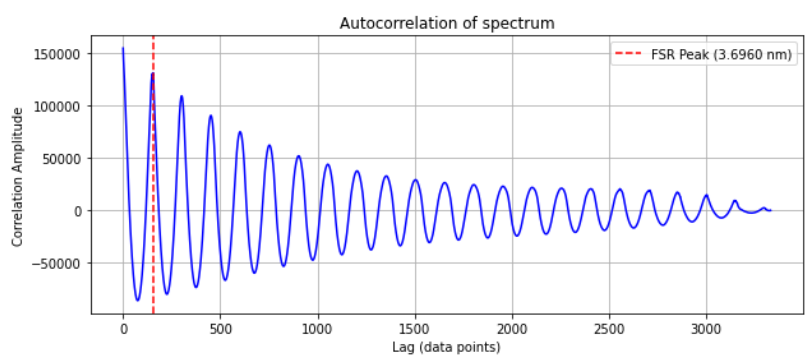
Fig. 11. Full analysis of MZI Layout 2. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4387$, $n_2 = -1.0999\,\mu m^{-1}$, and $n_3 = 0.0061\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.



(a) Experimental data vs. fitted MZI transfer function for Layout 3.

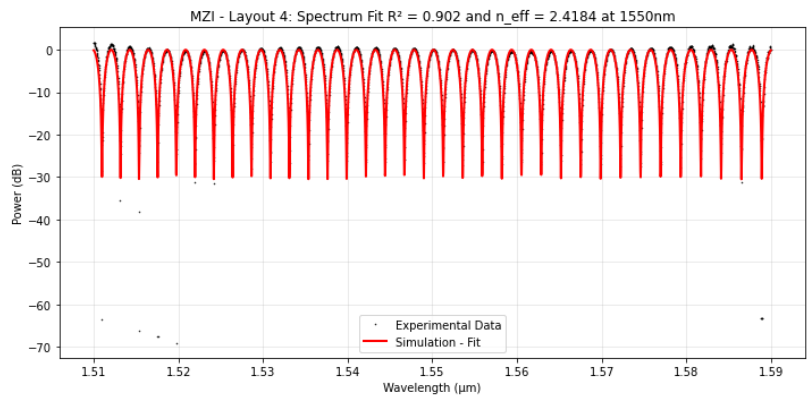


(b) Extracted group index n_g for Layout 3.

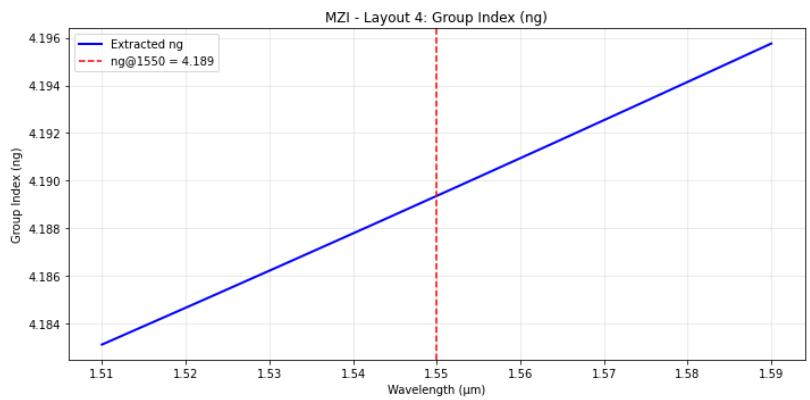


(c) Autocorrelation for Layout 3 used to determine initial FSR.

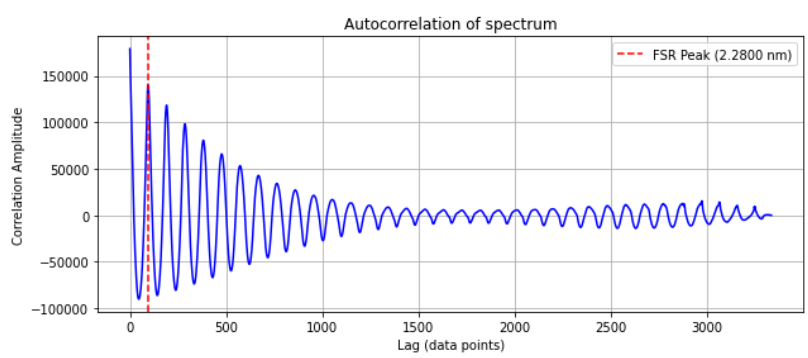
Fig. 12. Full analysis of MZI Layout 3. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4444$, $n_2 = -1.1104\,\mu m^{-1}$, and $n_3 = -0.0114\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.



(a) Experimental data vs. fitted MZI transfer function for Layout 4.

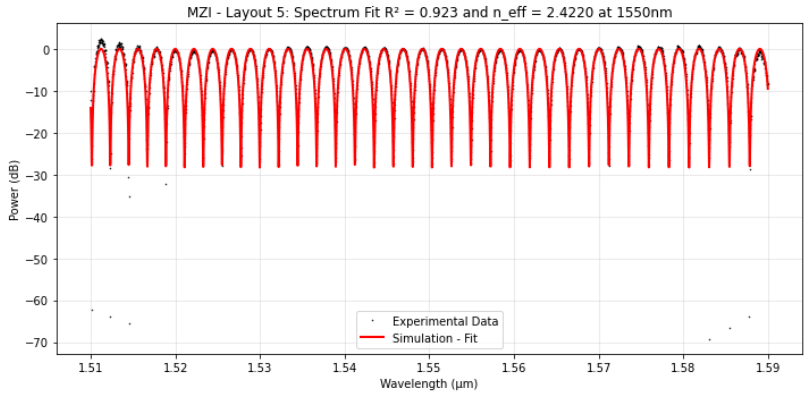


(b) Extracted group index n_g for Layout 4.

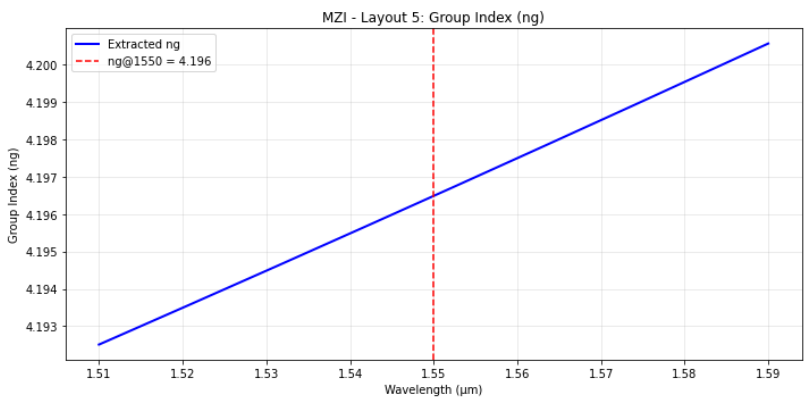


(c) Autocorrelation for Layout 4 used to determine initial FSR.

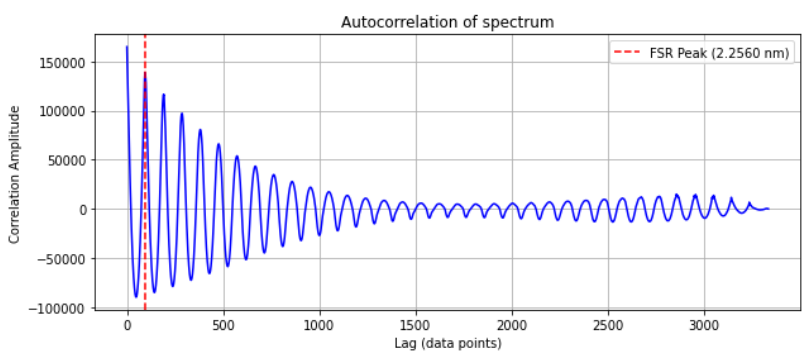
Fig. 13. Full analysis of MZI Layout 4. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4184$, $n_2 = -1.1426\,\mu m^{-1}$, and $n_3 = -0.0510\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.



(a) Experimental data vs. fitted MZI transfer function for Layout 5.



(b) Extracted group index n_g for Layout 5.



(c) Autocorrelation for Layout 5 used to determine initial FSR.

Fig. 14. Full analysis of MZI Layout 5. The effective index is modeled as $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$, with extracted parameters: $n_1 = 2.4220$, $n_2 = -1.1448\,\mu m^{-1}$, and $n_3 = -0.0325\,\mu m^{-2}$. (a) Final numerical fit, (b) Group index extraction, and (c) Initial peak detection.

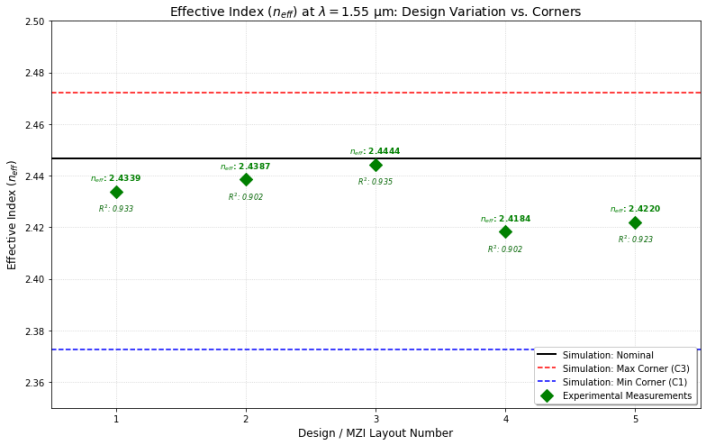


Fig. 15. Effective index (n_{eff}) at 1.55 µm. Comparison between experimental results (diamonds) and simulation corner values.

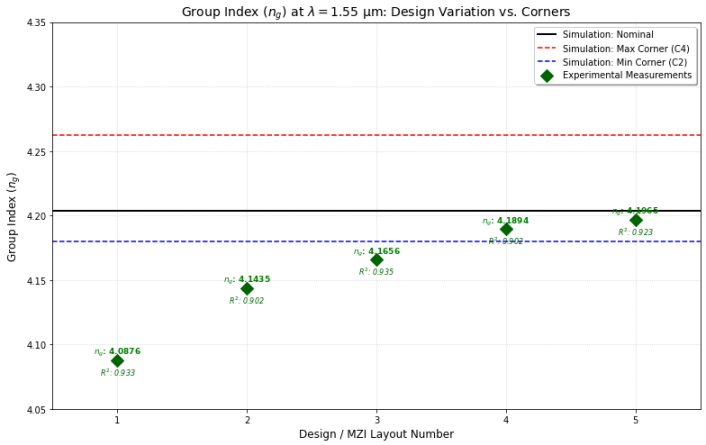


Fig. 16. Group index (n_g) at 1.55 µm. Comparison between experimental (diamonds) and simulation corner values.

Table 6. Comparison of Predicted vs. Experimental FSR (nm) across Manufacturing Corners for MZI Designs 1–5. Error (%) is calculated as $(FSR_{exp} - FSR_{nom})/FSR_{nom} \times 100$.

Design #	ΔL **(µm)**	**Nominal FSR**	**Min FSR (C2)**	**Max FSR (C4)**	**Exp. FSR (nm)**	**Error (%)**
1	40	14.289	14.092	14.370	**14.298**	+0.06%
2	80	7.144	7.046	7.185	**7.181**	+0.52%
3	160	3.572	3.523	3.593	**3.579**	+0.20%
4	250	2.286	2.255	2.299	**2.308**	+0.96%
5	250	2.286	2.255	2.299	**2.301**	+0.66%

For design 6 (TM mode), it was not possible to extract its experimental values due to fabrication issues.

Analysis of the extracted effective index (n_{eff}) and group index (n_g) show a good match with the simulated values. The extracted Free Spectral Range (FSR) parameter, across all TE mode layouts, followed the expected $1/\Delta L$ relationship, varying approximately from 14.3 nm to 2.3 nm.

The calculated error in FSR, which was below 1% for all MZI designs, suggests that the fabricated waveguides have their cross-section a bit narrower or thinner than the nominal 500×220 nm design, considered in this study. By consequence, it will be identified deviations in the group index (n_g), affecting the FSR directly.

Finally, the comparison between the straight geometry of Layout 4 and the serpentine format of Layout 5 demonstrates that folding does not compromise performance, thus they can be used to save space in the silicon area. Both designs shared similar results.

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